

Data Science Life Cycle in Healthcare

Introduction

Data Science is used in healthcare to analyze patient data and help doctors make better decisions. One common application is predicting whether a patient is at risk of developing heart disease based on factors such as age, blood pressure, cholesterol levels, and lifestyle habits.

The Data Science Life Cycle consists of six important stages:

- Data Collection
- Data Cleaning
- Exploratory Data Analysis (EDA)
- Modeling
- Testing
- Monitoring

1. Data Collection

Definition

Data collection is the process of gathering relevant information from various sources.

Example

For a heart disease prediction system, data can be collected from:

- Hospital records
- Laboratory reports
- Patient surveys
- Wearable devices (smartwatches, fitness bands)

Sample Data

- Patient ID
- Age
- BP
- Cholesterol
- Heart Disease

To gather enough patient information for analysis and prediction.

2. Data Cleaning

Definition

Data cleaning removes errors, missing values, and duplicate records from the collected data.

Example

Before Cleaning:

Age 220

BP NULL

Cholesterol NULL

Cleaning Activities

- Remove duplicate patient records
- Fill missing values
- Correct data entry errors
- Standardize formats

Purpose

To improve data quality and accuracy.

3. Exploratory Data Analysis (EDA)

Definition

EDA helps understand patterns, trends, and relationships in the data.

Example

Questions explored:

- Does age affect heart disease risk?
- Is high cholesterol related to heart disease?
- Which factor contributes most to disease prediction?

Findings

- Patients above 50 years have higher risk.
- High cholesterol levels increase disease probability.
- High blood pressure is strongly associated with heart disease.

Common Visualizations

- Bar Charts
- Histograms
- Pie Charts
- Scatter Plots

Purpose

To discover useful insights before building a model.

4. Modeling

Definition

Modeling involves training a machine learning algorithm using historical data.

Example

Input Features:

1. Age
2. Blood Pressure
3. Cholesterol
4. Smoking Habit
5. Exercise Frequency

Output:

1. Heart Disease (Yes/No)
2. Algorithms Used
3. Logistic Regression
4. Decision Tree

Example

Input:

- Age = 55
- BP = 145
- Cholesterol = 240

Prediction:

Heart Disease Risk = High

Purpose

To make accurate predictions for new patients.

5. Testing

Definition

Testing evaluates the model's performance on unseen data.

Example

The model is tested using patient records that were not used during training.

Evaluation Metrics

- Accuracy
- Precision

Sample Result

- Accuracy 92%
- Precision 90%

Purpose

To ensure the model gives reliable predictions.

6. Monitoring

Definition

Monitoring continuously checks the model's performance after deployment.

Example

After deployment in hospitals:

- Track prediction accuracy
- Detect changes in patient data patterns
- Update the model when needed

Example

If a new disease pattern emerges and prediction accuracy drops from 92% to 80%, the model is retrained with new patient data.

Purpose

To maintain accuracy and effectiveness over time